

Proposed Solution

Date	11 Jun 2026
Team ID	
Project Name	OptiCrop
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Farmers encounter volatile crop yields due to unpredictable soil nutritional trends and manual reliance on traditional guesswork, lacking immediate mathematical tools to assess land viability beforehand.
2.	Idea / Solution description	OptiCrop builds an intelligent data-driven portal that accepts key parameters like Nitrogen (N), Phosphorus (P), Potassium (K), and pH levels to instantly run a Random Forest multiclass predictive pipeline, serving precision crop recommendations.
3.	Novelty / Uniqueness	Utilizes lightweight localized classification matrices to deliver micro-seconds evaluation execution offline, bypassing commercial laboratory delay walls without expensive infrastructure.
4.	Social Impact / Customer Satisfaction	Minimizes agricultural capital input waste risks for rural farmlands, transforming stress points into confident production planning metrics for local cooperative communities.
5.	Business Model (Revenue Model)	Freemium core usage tiers for local family growers, with subscription validation matrices and custom API integration points for B2B agricultural consulting organizations and regional fertilizer suppliers.
6.	Scalability of the Solution	Completely stateless, modular microservices architecture layout that allows scaling background ML matrix endpoints instantly via localized containers without altering front-end portals.

